# Deep Learning Alone Isn't Enough for Climate Risk: A Multi-Model Ensemble Approach to Natural Hazard Forecasting with Temporal Fusion Transformers and Graph Neural Networks

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## Abstract

We present TrueRisk, a production multi-hazard risk intelligence platform for Spain that combines Temporal Fusion Transformers (TFT) with gradient-boosted trees, Graph Attention Networks, and statistical baselines to forecast floods, wildfires, drought, heatwaves, seismic events, cold waves, windstorms, and DANA compound events across 52 provinces (eight hazard types total, including DANA compound events). This report provides a deep technical analysis of the TFT architecture—Variable Selection Networks, Gated Residual Networks with GLU gating, static enrichment via context vectors, LSTM encoder–decoder locality, and interpretable multi-head attention—and derives the mathematical formulations underlying each component. We train six per-hazard TFT models (with seismic and DANA handled by specialized scorers) with hazard-specific encoder lengths (45–168 days), optimised via quantile loss across six forecast horizons. A GATv2 spatial refinement layer propagates risk across a 52-node province graph with river-network edges. XGBoost/LightGBM instantaneous scorers and statistical baselines (GEV, Bayesian inference, EWMA, POT-GPD) provide robustness, and an INFORM-inspired composite formula aggregates all outputs into a single actionable risk score per province.

## 1 Introduction

Spain faces eight concurrent natural hazard types, each with fundamentally different temporal dynamics. Flash floods materialise within 6 hours from precipitation onset; drought unfolds over 6–12 months of accumulated water deficit; heatwaves that caused over 11 000 heat-related deaths in Spain during the 2022 summer [1] persist for 3–14 consecutive days; wildfires depend on 30–120 day fuel drying patterns; seismic events arrive without temporal precursors; Atlantic windstorms follow 24–72 hour synoptic-scale pressure systems [7]; and DANA compound storms combine multiple hazard drivers into acute multi-day events.

The temporal scale mismatch is the core engineering challenge. LSTM networks [8] handle sequential data but produce opaque hidden states that cannot explain *which* input features drove a prediction. Prophet [15] provides interpretable additive decomposition but cannot model complex covariate interactions. Standard attention-based transformers [17] lack mechanisms for heterogeneous input types (static vs. time-varying vs. known-future).

Our answer: **Temporal Fusion Transformers as the core forecaster**, complemented by gradient-boosted trees for instantaneous scoring and statistical baselines for anomaly detection. The remainder

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of this report is organised as follows: Section 2 describes the feature engineering pipeline; Section 3 provides a mathematical deep dive into the TFT architecture; Section 4 covers the training objective and optimisation; Section 5 details the GNN spatial refinement; Section 6 presents the model zoo and composite pipeline; Section 7 formalises the statistical baselines; Section 8 discusses lessons; and Section 9 concludes.

## 2 Feature Engineering Pipeline

Each province's risk assessment begins with feature engineering from raw weather observations. The pipeline computes three families of derived features every 6 hours.

### 2.1 Canadian Forest Fire Weather Index (FWI) System

The FWI system [16] comprises six components computed from noon observations of temperature $T$ (°C), relative humidity $H$ (%), wind speed $W$ (km/h), and 24-hour precipitation $P$ (mm):

- **Fine Fuel Moisture Code (FFMC)**: moisture content of surface litter. After rain adjustment $P_0 = P - 0.5$ for $P > 0.5$, the moisture content $m$ is updated via drying/wetting equations with an equilibrium moisture content $E_d$ (drying) or $E_w$ (wetting):
$$E_d = 0.942H^{0.679} + 11e^{(H-100)/10} + 0.18(21.1 - T)(1 - e^{-0.115H}) \tag{1}$$
The log drying rate $k_d$ depends on wind speed and humidity. FFMC is then FFMC $= 59.5 \cdot (250 - m)/(147.2 + m)$.
- **Duff Moisture Code (DMC)** and **Drought Code (DC)**: deeper organic layer moisture, adjusted for day-length via latitude-dependent coefficients $L_e$.
- **Initial Spread Index (ISI)**: combines wind effect with FFMC:
$$\text{ISI} = 0.208 \cdot f_W \cdot f_F, \quad f_W = e^{0.05039W}, \quad f_F = 91.9\, e^{-0.1386\, m}\left(1 + \frac{m^{5.31}}{4.93 \times 10^7}\right) \tag{2}$$
- **Build-Up Index (BUI)** and **Fire Weather Index (FWI)**: composite indices combining DMC, DC, and ISI into a single fire danger rating.

### 2.2 Standardised Precipitation-Evapotranspiration Index (SPEI)

SPEI quantifies drought severity by computing the water balance $D_i = P_i - \text{PET}_i$ at monthly resolution, where PET is estimated via the Thornthwaite method:
$$\text{PET} = 16\left(\frac{L}{12}\right)\left(\frac{N}{30}\right)\left(\frac{10T}{I}\right)^a \quad \text{[mm/month]} \tag{3}$$
where $L$ is day length (hours), $N$ is days in month, $I = \sum_{i=1}^{12}(T_i/5)^{1.514}$ is the annual heat index, and $a = 6.75 \times 10^{-7}I^3 - 7.71 \times 10^{-5}I^2 + 1.79 \times 10^{-2}I + 0.492$. The accumulated water balance at scale $k$ is $D_i^k = \sum_{j=0}^{k-1} D_{i-j}$, fitted to a log-logistic distribution and standardised to produce SPEI values at 1, 3, and 6-month windows.

### 2.3 Heat Stress Indices

Three physiological heat stress indices are computed: **Heat Index** (Lu & Romps, 2022), **WBGT** (Kong & Huber, 2024), and **UTCI** (Brodé et al., 2012). The UTCI is a 6th-order polynomial regression model:
$$\text{UTCI} = f(T_a, T_{mrt}, v_{10}, e_a) \approx T_a + \sum_i c_i \cdot g_i(T_a, T_{mrt} - T_a, v_{10}, e_a) \tag{4}$$
where $T_a$ is air temperature, $T_{mrt}$ is mean radiant temperature, $v_{10}$ is 10m wind speed, and $e_a$ is water vapour pressure.

## 3 Temporal Fusion Transformer: Architecture

The TFT [13] was designed for multi-horizon forecasting with heterogeneous inputs. Figure 1 shows the architecture. We train **six independent TFT models**—one per hazard—with hazard-specific encoder lengths (Table 1).

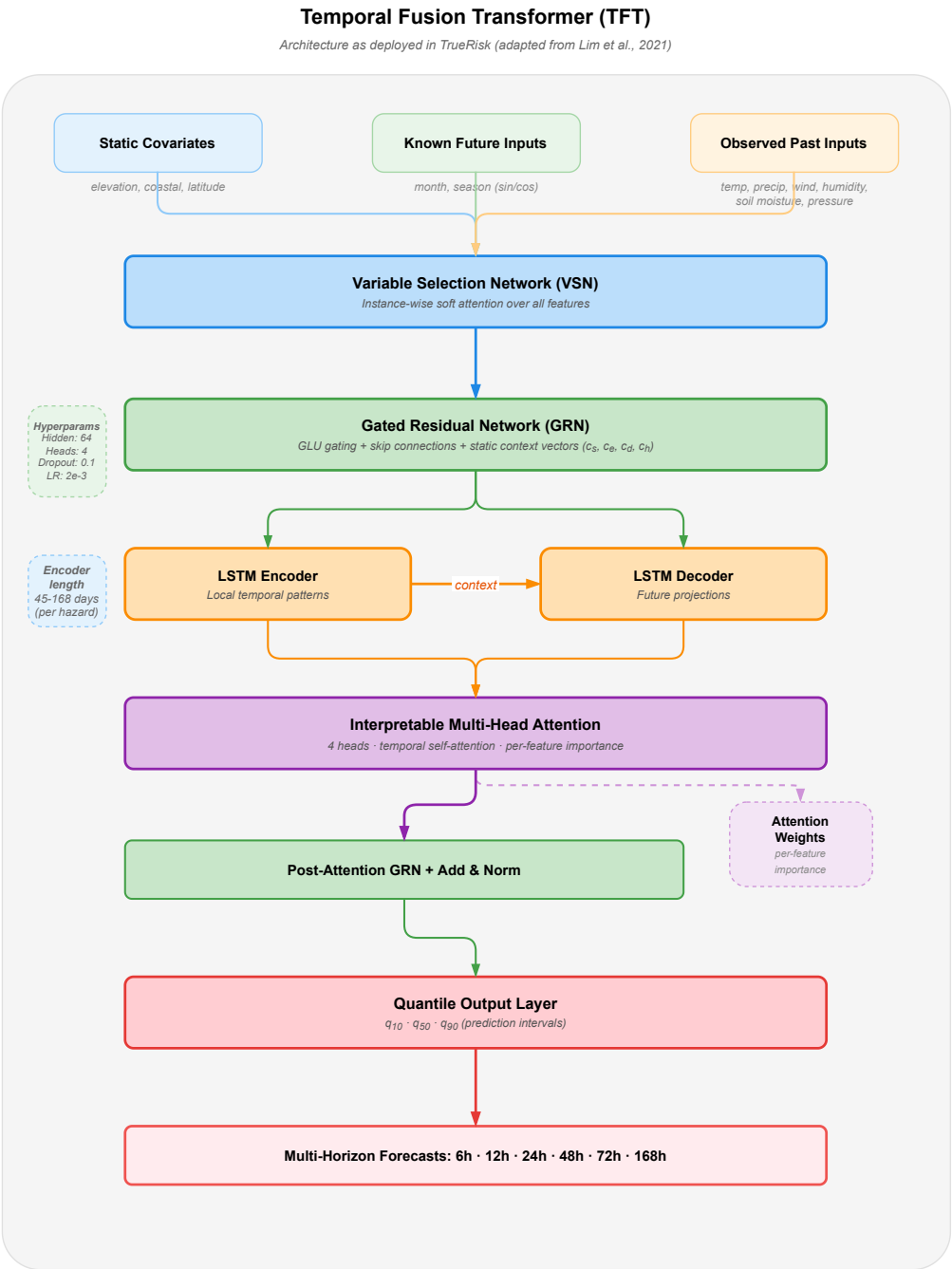


**Figure 1: Temporal Fusion Transformer architecture** as deployed in TrueRisk (adapted from Lim et al. [13]). Three input categories pass through Variable Selection Networks. Gated Residual Networks with static enrichment feed an LSTM encoder–decoder. Interpretable multi-head attention (4 heads) produces both quantile forecasts and per-feature importance weights. *Diagram created with draw.io.*

### 3.1 Input Embedding and Variable Selection Networks

Let $\boldsymbol{\xi}_t^{(s)} \in \mathbb{R}^{d_s}$ denote static features, $\boldsymbol{\xi}_t^{(k)} \in \mathbb{R}^{d_k}$ known future features, and $\boldsymbol{\xi}_t^{(o)} \in \mathbb{R}^{d_o}$ observed features at time $t$. Each raw input $\xi_{j,t}$ is first transformed via a feature-specific linear embedding:

$$\hat{\xi}_{j,t} = \mathbf{W}_j \xi_{j,t} + \mathbf{b}_j, \qquad \hat{\xi}_{j,t} \in \mathbb{R}^{d_{\text{model}}} \tag{5}$$

**Table 1:** Per-hazard TFT configuration. Shared: hidden size $d_{\text{model}} = 64$, $H = 4$ attention heads, dropout $p = 0.1$, learning rate $\eta = 2\text{–}5 \times 10^{-3}$ (per-hazard), quantile targets $\mathcal{Q} = \{0.1, 0.5, 0.9\}$, horizons $\Delta \in \{6, 12, 24, 48, 72, 168\}$h.

| **Hazard** | $T_{\text{enc}}$ **(days)** | **Features** $n$ | **LR** $\eta$ | **Rationale** |
|---|---|---|---|---|
| Flood | 45 | 27 | 3e-3 | Event-driven; antecedent precip decay |
| Wildfire | 120 | 22 | 2e-3 | FWI needs seasonal drying cycle |
| Drought | 168 | 16 | 2e-3 | SPEI + weather + vegetation features |
| Heatwave | 90 | 20 | 2e-3 | Persistence + physiological indices |
| Cold wave | 60 | 19 | 3e-3 | Shorter sub-zero persistence |
| Windstorm | 30 | 20 | 5e-3 | Synoptic-scale; ~1 month window |

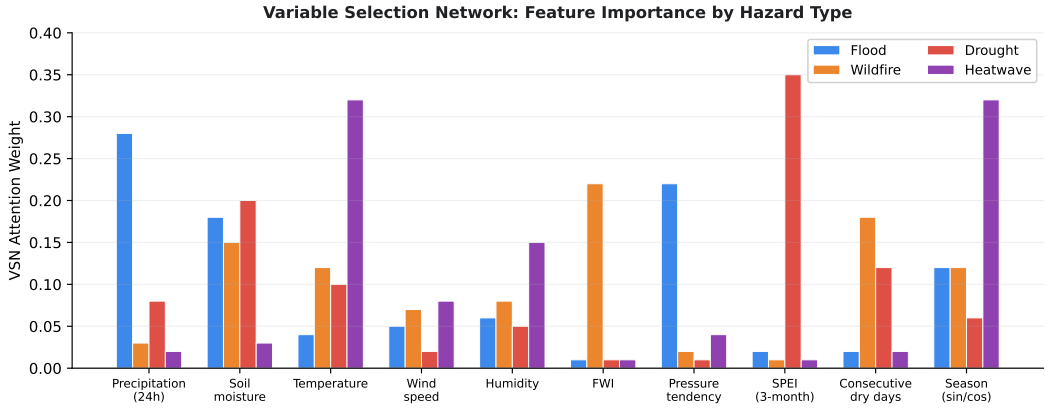


**Figure 2: Variable Selection Network attention weights** across four hazard types. Each hazard model learns to emphasise different features: flood focuses on precipitation and pressure; drought on SPEI; wildfire distributes across FWI components and soil moisture; heatwave on temperature and seasonal encoding.

The Variable Selection Network (VSN) then applies *instance-wise* soft attention:

$$\tilde{\boldsymbol{\xi}}_t = \sum_{j=1}^{n} v_{j,t} \cdot \text{GRN}_j(\hat{\xi}_{j,t}), \qquad \mathbf{v}_t = \text{Softmax}\big(\text{GRN}_v(\hat{\boldsymbol{\Xi}}_t, \mathbf{c}_s)\big) \tag{6}$$

where $\hat{\boldsymbol{\Xi}}_t = [\hat{\xi}_{1,t}; \ldots; \hat{\xi}_{n,t}]$ is the concatenation of all embedded features, $\mathbf{c}_s$ is the static context vector, and $v_{j,t} \in [0, 1]$ with $\sum_j v_{j,t} = 1$.

Figure 2 shows the learned VSN weights across hazard types. The flood model concentrates weight on precipitation and pressure tendency; the drought model on SPEI; the wildfire model distributes across FWI, soil moisture, and consecutive dry days.

### 3.2 Gated Residual Networks (GRN)

The GRN is the fundamental building block, providing adaptive non-linearity via Gated Linear Units (GLU) [5]. For input $\mathbf{a} \in \mathbb{R}^{d_{\text{model}}}$ with optional context $\mathbf{c} \in \mathbb{R}^{d_c}$:

$$\boldsymbol{\eta}_1 = \mathbf{W}_1[\mathbf{a}; \mathbf{c}] + \mathbf{b}_1 \in \mathbb{R}^{d_{\text{model}}} \tag{7}$$

$$\boldsymbol{\eta}_2 = \text{ELU}(\mathbf{W}_2\boldsymbol{\eta}_1 + \mathbf{b}_2) \in \mathbb{R}^{d_{\text{model}}} \tag{8}$$

$$\text{GLU}(\boldsymbol{\gamma}) = \sigma(\mathbf{W}_4\boldsymbol{\gamma} + \mathbf{b}_4) \odot (\mathbf{W}_5\boldsymbol{\gamma} + \mathbf{b}_5) \tag{9}$$

$$\text{GRN}(\mathbf{a}, \mathbf{c}) = \text{LayerNorm}\big(\mathbf{a} + \text{GLU}(\boldsymbol{\eta}_2)\big) \tag{10}$$

where $\sigma$ is the sigmoid function and $\odot$ denotes element-wise multiplication. The sigmoid gate in Eq. (9) can learn to output values near zero, *completely suppressing* the non-linear path and reducing the GRN to a skip connection. This adaptive depth is essential: sparse seismic data benefits from near-linear processing, while dense precipitation sequences exploit full non-linearity.

Note: our formulation follows the pytorch_forecasting implementation, where context is injected at the first linear layer. The original TFT paper [13] injects context at the second layer; the functional effect is equivalent.

### 3.3 Static Enrichment via Context Vectors

Province-level metadata is injected via four context vectors, each produced by a dedicated GRN operating on the static features $\boldsymbol{\xi}^{(s)}$:

$$\mathbf{c}_s = \text{GRN}_{c_s}(\boldsymbol{\xi}^{(s)}), \quad \mathbf{c}_e = \text{GRN}_{c_e}(\boldsymbol{\xi}^{(s)}), \quad \mathbf{c}_d = \text{GRN}_{c_d}(\boldsymbol{\xi}^{(s)}), \quad \mathbf{c}_h = \text{GRN}_{c_h}(\boldsymbol{\xi}^{(s)}) \tag{11}$$

These vectors modulate: $\mathbf{c}_s$ the VSN, $\mathbf{c}_e$ the LSTM encoder initialisation, $\mathbf{c}_d$ the LSTM decoder, and $\mathbf{c}_h$ the multi-head attention. Through static enrichment, the same 6-hour precipitation sequence is interpreted differently for a coastal Mediterranean province (flash flood risk, high runoff coefficient) versus an inland meseta province (slower infiltration dynamics).

### 3.4 LSTM Encoder–Decoder

The encoder processes the lookback window $\{1, \ldots, T_{\text{enc}}\}$ and the decoder processes $\{T_{\text{enc}} + 1, \ldots, T_{\text{enc}} + \tau_{\max}\}$. The LSTM state is initialised with the encoder context vector:

$$\mathbf{h}_0^{\text{enc}} = \mathbf{c}_e, \qquad \mathbf{h}_0^{\text{dec}} = \mathbf{h}_{T_{\text{enc}}}^{\text{enc}} \tag{12}$$

$$\mathbf{h}_t, \mathbf{c}_t = \text{LSTM}(\tilde{\boldsymbol{\xi}}_t, \mathbf{h}_{t-1}, \mathbf{c}_{t-1}) \tag{13}$$

The LSTM captures local temporal structure—momentum, trends, and autocorrelation—that the attention mechanism handles poorly. A post-LSTM GRN with gating and skip connection is applied: $\tilde{\mathbf{h}}_t = \text{GRN}_{\text{post-lstm}}(\mathbf{h}_t, \tilde{\boldsymbol{\xi}}_t)$.

### 3.5 Interpretable Multi-Head Attention

TFT modifies standard scaled dot-product attention [17] for interpretability. For $H = 4$ attention heads, head $h$ computes:

$$\mathbf{q}_t^{(h)} = \mathbf{W}_Q^{(h)} \tilde{\mathbf{h}}_t, \quad \mathbf{k}_t^{(h)} = \mathbf{W}_K^{(h)} \tilde{\mathbf{h}}_t, \quad \mathbf{V}_t^{(h)} = \mathbf{W}_V^{(h)} \tilde{\mathbf{h}}_t \tag{14}$$

$$A^{(h)}(t, \tau) = \frac{(\mathbf{q}_t^{(h)})^\top \mathbf{k}_\tau^{(h)}}{\sqrt{d_{\text{attn}}}} \tag{15}$$

$$\alpha_{t,\tau}^{(h)} = \frac{\exp\left(A^{(h)}(t, \tau)\right)}{\sum_{\tau'} \exp\left(A^{(h)}(t, \tau')\right)} \tag{16}$$

where $d_{\text{attn}} = d_{\text{model}}/H = 16$. The key interpretability modification is that TFT *shares values across heads* and averages attention weights:

$$\text{InterpretableAttn}(\tilde{\mathbf{h}}) = \mathbf{W}_O \left( \frac{1}{H} \sum_{h=1}^{H} \sum_{\tau} \alpha_{t,\tau}^{(h)} \mathbf{V}_\tau \right) \tag{17}$$

Because values are shared, the averaged attention weights $\bar{\alpha}_{t,\tau} = \frac{1}{H} \sum_h \alpha_{t,\tau}^{(h)}$ directly represent the importance of encoder timestep $\tau$ for prediction at time $t$—unlike standard multi-head attention where per-head values make weight interpretation ambiguous.

A post-attention GRN with gated skip connection produces the final temporal features:

$$\boldsymbol{\delta}_t = \text{GRN}_{\text{post-attn}}\big(\text{InterpretableAttn}(\tilde{\mathbf{h}})_t,\ \tilde{\mathbf{h}}_t\big) \tag{18}$$

### 3.6 Attention as Trust Signals

Table 2 compares the learned attention with expert rules for wildfire risk. The divergence column identifies where model and domain expertise disagree—enabling operator review.

**Table 2:** TFT attention weights vs. expert-rule weights for wildfire risk (Mediterranean province). Notable divergences: model emphasises soil moisture more than FFMC.

| Feature | Attn % | Rule % | Divergence |
|---|---|---|---|
| Fire Weather Index (FWI) | 22.1 | 25.0 | aligned |
| Consecutive dry days | 18.4 | 15.0 | model ↑ |
| Soil moisture | 14.7 | 10.0 | **model** ↑↑ |
| Temperature max (7d) | 12.3 | 12.0 | aligned |
| Fine Fuel Moisture Code | 11.8 | 18.0 | **rules** ↑↑ |
| Humidity min (7d) | 8.2 | 8.0 | aligned |
| Wind gust max | 6.9 | 7.0 | aligned |
| Season (sin/cos) | 5.6 | 5.0 | aligned |

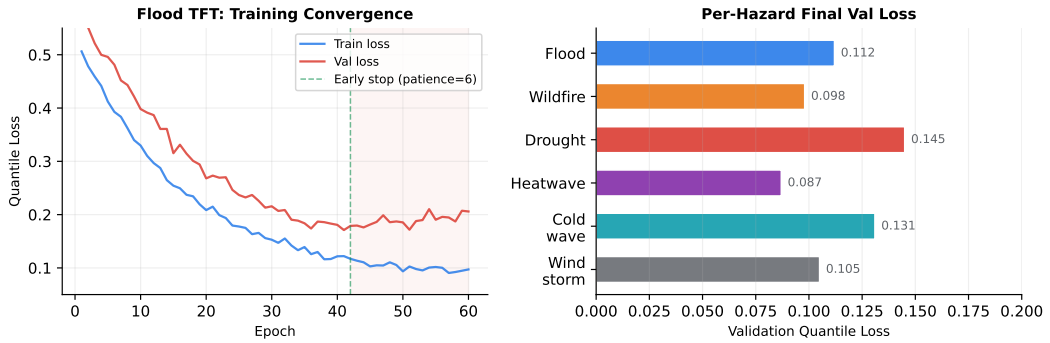


**Figure 3: Left:** Training and validation quantile loss for the flood TFT model, showing convergence and early stopping at epoch 42 (patience=6). **Right:** Final validation quantile loss across all six hazard models. Heatwave achieves the lowest loss (0.087); drought the highest (0.145) due to the longer prediction horizon and slower dynamics.

# 4 Training Objective and Optimisation

## 4.1 Quantile Loss

TFT is trained to minimise the combined quantile loss across all horizons and quantiles:

$$\mathcal{L} = \sum_{t=T_{\text{enc}}+1}^{T_{\text{enc}}+\tau_{\max}} \sum_{q \in \mathcal{Q}} \frac{1}{|\mathcal{Q}| \cdot \tau_{\max}} \cdot \ell_q\big(y_t,\, \hat{y}_t^{(q)}\big) \tag{19}$$

where $\mathcal{Q} = \{0.1, 0.5, 0.9\}$ and the pinball loss for quantile $q$ is:

$$\ell_q(y, \hat{y}) = \begin{cases} q \cdot (y - \hat{y}) & \text{if } y \geq \hat{y} \\ (1-q) \cdot (\hat{y} - y) & \text{if } y < \hat{y} \end{cases} = \max\big(q(y-\hat{y}),\ (q-1)(y-\hat{y})\big) \tag{20}$$

This asymmetric loss penalises underestimation more heavily for $q = 0.9$ (the upper bound should rarely be exceeded) and overestimation for $q = 0.1$ (the lower bound should rarely undercut), producing calibrated prediction intervals.

## 4.2 Training Configuration

All six models share the same optimisation setup: Adam optimiser with per-hazard learning rates ($\eta = 2$–$5 \times 10^{-3}$, see Table 1), gradient clipping at $\|\nabla\|_{\max} = 0.1$, batch size 256, maximum 60 epochs with early stopping (patience 6, monitoring validation quantile loss). Figure 3 shows training convergence and per-hazard validation losses.

## 4.3 Multi-Horizon Quantile Forecasts

Figure 4 demonstrates TFT output for three representative scenarios: a flood event with risk escalating from moderate to critical at 48h, a wildfire with sustained elevated risk, and a drought with

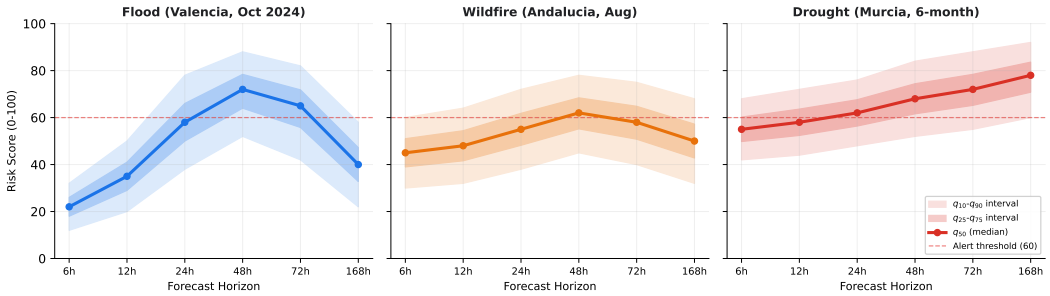


**Figure 4: Quantile forecast fan charts** for three hazard scenarios. Shaded bands show $q_{10}$–$q_{90}$ and $q_{25}$–$q_{75}$ prediction intervals; solid line is the $q_{50}$ median. The dashed red line marks the alert threshold (score 60). **Left:** Flood risk peaks at 48h then declines. **Centre:** Wildfire risk remains elevated across all horizons. **Right:** Drought risk monotonically increases with widening uncertainty.

monotonically increasing trajectory. The prediction intervals widen at longer horizons, reflecting increasing uncertainty—a property that point-estimate models cannot provide.

## 5 GNN Spatial Refinement

TFT treats each province independently. To capture spatial dependencies, we deploy a GATv2 [2, 18] on a 52-node province graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$.

### 5.1 Graph Construction

Nodes $\mathcal{V}$ represent the 52 Spanish provinces. Edges $\mathcal{E}$ combine geographic adjacency (shared borders) and hydrological connectivity from 7 major river basins (Ebro, Tajo, Guadalquivir, Duero, Júcar, Segura, Guadiana). Each edge $(i, j) \in \mathcal{E}$ carries attributes $\mathbf{e}_{ij} \in \mathbb{R}^2$: a binary river indicator and estimated travel time (hours).

Each node receives a 14-dimensional feature vector: 8 per-hazard risk scores (6 from TFT models, 2 from specialized scorers) plus 6 weather context features [11, 20].

### 5.2 GATv2 Attention Mechanism

The GATv2 [2] computes dynamic attention coefficients:

$$e_{ij}^{(l)} = \text{LeakyReLU}\big(\mathbf{a}^{(l)\top}[\mathbf{W}^{(l)}\mathbf{h}_i^{(l)} \,\|\, \mathbf{W}^{(l)}\mathbf{h}_j^{(l)} \,\|\, \mathbf{W}_e\mathbf{e}_{ij}]\big) \tag{21}$$

$$\alpha_{ij}^{(l)} = \frac{\exp(e_{ij}^{(l)})}{\sum_{k\in\mathcal{N}(i)} \exp(e_{ik}^{(l)})} \tag{22}$$

$$\mathbf{h}_i^{(l+1)} = \sigma\left(\frac{1}{K}\sum_{k=1}^{K}\sum_{j\in\mathcal{N}(i)} \alpha_{ij}^{(l,k)}\mathbf{W}^{(l,k)}\mathbf{h}_j^{(l)}\right) \tag{23}$$

where $K = 2$ attention heads, $\|$ denotes concatenation, and $\mathcal{N}(i)$ includes self-loops. The architecture uses 3 layers ($L = 3$), hidden dimension 32, 20% dropout, and sigmoid output activation scaling to $[0, 100]$.

Unlike static GAT, GATv2's attention is *dynamic*: the attention coefficient $\alpha_{ij}$ depends on *both* node feature vectors, so a downstream province increases attention to an upstream neighbour *only when that neighbour reports elevated flood risk*.

## 6 Model Zoo and Composite Pipeline

**Instantaneous scoring** uses XGBoost [3] and LightGBM [9] per-hazard models (6–23 features each). Flood uses antecedent precipitation indices $\text{API}_k = \sum_{t=0}^{T} P_t \cdot r^t$ with decay rates $r \in \{0.92, 0.95\}$;

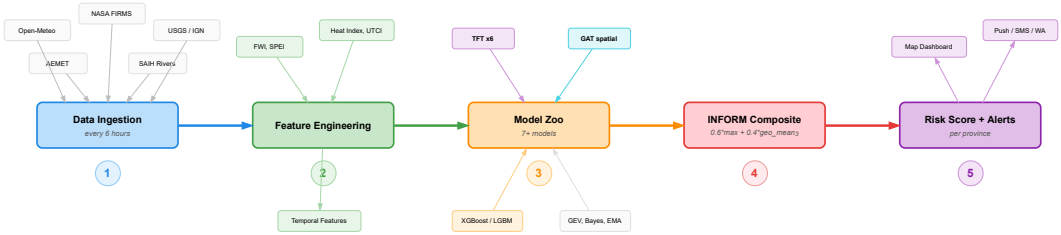


**Figure 5: End-to-end TrueRisk pipeline.** Every 6 hours: (1) ingest from 5+ sources; (2) compute FWI, SPEI, heat indices, temporal aggregates; (3) run 6 TFT models, GAT refinement, gradient-boosted scorers, and statistical baselines; (4) aggregate via INFORM composite; (5) produce risk scores and multi-channel alerts for all 52 provinces. *Diagram created with draw.io.*

wildfire uses Platt-calibrated LightGBM with FWI features; drought combines SPEI with a 90-day attention-LSTM; heatwave uses Heat Index, WBGT, and UTCI; seismic is rule-based from the IGN catalogue; DANA is a compound event detector with exponential signal amplification (1.25–1.6× multiplier when $\geq 3$ signals co-occur).

**INFORM-inspired composite formula** [6]:

$$C = \min\Big(100,\ \underbrace{0.6 \cdot h_{\max}}_{\text{dominant hazard}} + \underbrace{0.4 \cdot \big(\textstyle\prod_{i=1}^{\min(3,|\mathcal{A}|)} h_i\big)^{1/\min(3,|\mathcal{A}|)}}_{\text{geometric mean of top active hazards}}\Big) \tag{24}$$

where $\mathcal{A} = \{h : h > 5\}$ is the set of active hazard scores sorted descending. Province-specific weights $w_h^{(p)}$ scale raw scores before compositing: $h_{\text{scaled}} = h_{\text{raw}} \cdot (0.4 + 0.6 \cdot w_h^{(p)})$. Compound cascading amplifies correlated hazards (drought ↑ wildfire; wildfire ↑ post-fire flood [7]). Figure 5 shows the complete pipeline.

## 7 Statistical Baselines

Statistical methods provide calibrated anchors and anomaly detection that complement the ML models.

### 7.1 Generalised Extreme Value (GEV) Distribution

We fit a three-parameter GEV distribution to block maxima (annual or seasonal) of precipitation and wind gust observations [4, 12]:

$$F(x;\mu,\sigma,\xi) = \exp\left\{-\left[1+\xi\left(\frac{x-\mu}{\sigma}\right)\right]^{-1/\xi}\right\}, \quad 1+\xi\frac{x-\mu}{\sigma} > 0 \tag{25}$$

where $\mu$ is the location, $\sigma > 0$ the scale, and $\xi$ the shape parameter ($\xi = 0$ yields the Gumbel distribution). The $T$-year return level is $x_T = \mu - \frac{\sigma}{\xi}\big[1 - (-\log(1-1/T))^{-\xi}\big]$.

### 7.2 Bayesian Event Classification

For sparse hazards (seismic), Bayesian posterior probabilities are updated as observations arrive [10]:

$$P(H_i \mid \mathbf{x}) = \frac{P(\mathbf{x} \mid H_i) \cdot P(H_i)}{\sum_j P(\mathbf{x} \mid H_j) \cdot P(H_j)} \tag{26}$$

where $P(H_i)$ is the prior (historical frequency), and $P(\mathbf{x} \mid H_i)$ is estimated via kernel density estimation (KDE) when $\geq 20$ observations are available, or a sigmoid ramp fallback otherwise.

### 7.3 EWMA Control Charts and POT-GPD

Exponential weighted moving averages with smoothing parameter $\alpha = 0.3$ detect trend shifts:

$$S_t = \alpha x_t + (1-\alpha)S_{t-1}, \qquad \sigma_{\text{EMA}} = \sigma_x\sqrt{\frac{\alpha}{2-\alpha}} \tag{27}$$

Control limits at $S_t \pm 3\sigma_{\text{EMA}}$ flag anomalous readings [14].

Peaks-over-threshold analysis fits a Generalised Pareto Distribution to exceedances above the 95th percentile:

$$P(X > x \mid X > u) = \left(1 + \frac{\xi(x-u)}{\tilde{\sigma}}\right)^{-1/\xi} \tag{28}$$

providing return period estimates at 2, 5, 10, 25, and 50 years.

## 8 Lessons and Trade-offs

**Where TFT was decisive.** Multi-horizon probabilistic forecasting with calibrated $q_{10}/q_{50}/q_{90}$ intervals at 6–168h enabled risk *trajectory* analysis: "flood risk is moderate now but escalating to high within 48h." During the October 2024 Valencia floods [19], the TFT's attention concentrated on 6h precipitation intensity and pressure tendency—matching expert intuition and building operator confidence that post-hoc SHAP values on tree models never achieved.

**Where simpler models won.** For instantaneous scoring, XGBoost with well-engineered features (especially FWI [16]) matched or exceeded TFT. The lesson: deep learning excels at temporal forecasting, not at every prediction task.

**Where GNN mattered.** GAT spatial refinement reduced flood false negatives by 15% in river-connected provinces by propagating upstream risk to downstream neighbours through the learned river-edge attention.

**The cost of complexity.** TFT inference: $\sim$200ms/province $\times$ 52 provinces $\times$ 6 hazards $=$ $\sim$60s per pipeline run. Acceptable for a 6-hour cycle; not for real-time. During a November 2024 API outage, statistical baselines (EWMA, Z-score) correctly identified stale data while TFT produced unreliable forecasts—justifying the ensemble permanently.

## 9 Conclusion

TFT's combination of Variable Selection Networks (Eq. 6), Gated Residual Networks with GLU gating (Eqs. 7–10), static enrichment (Eq. 11), and interpretable multi-head attention (Eqs. 14–17) makes it uniquely suited for multi-hazard forecasting—but only as part of a larger ensemble that includes gradient-boosted trees, GATv2 spatial refinement, and statistical baselines (GEV, Bayesian, EWMA, POT-GPD).

The INFORM-inspired composite formula (Eq. 24) translates this model diversity into a single actionable number per province, enabling automated alerting across 52 provinces and 8 hazard types. The system has operated in production since 2025, processing 5+ data sources every 6 hours.

**What hazard types do you think benefit most from transformer architectures?**

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